

Total No. of Questions : 8]

SEAT No. :

PC-1697

[Total No. of Pages :2

[63531-13

T.E. (CIVIL) (301014)

**Remote Sensing & Geographic Information System
(2019 PATTERN) (SEMESTER — II)**

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q5 or Q6, Q7 or Q8
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume Suitable data if necessary.
- 5) Use of electronic pocket calculator is allowed in the examination.

- Q1)** a) Give a brief account of Origin of Global positioning system. [6]
b) What are the three segments of GPS? Describe them briefly. [6]
c) Illustrate the advantages and disadvantages of GNSS. [5]

OR

- Q2)** a) Explain the concept of GNSS. [6]
b) Write a short note on following; i) GLONASS system ii) Galileo system. [6]
c) Describe the different sources of error in GPS. [5]

- Q3)** a) Differentiate between supervised and unsupervised image classification. [6]
b) State the application of DEM. [6]
c) Define Triangular Irregular Network Model (TIN) and its applications. [6]

OR

- Q4)** a) Write a note on digital image and analog image. [6]
b) Explain in detail the concept of post processing and pre processing. [6]
c) Differentiate between Visual and Digital image interpretation. [6]

P.T.O.

- Q5)** a) Explain the functionalities of GIS? [6]
b) Write in detail about various components of GIS. [6]
c) List down the differences between drafting software's and GIS. [5]

OR

- Q6)** a) Illustrate the Geographical Information System and state Spatial Data types. [6]
b) Demonstrate the link between remote sensing & GIS. [6]
c) Compare the various definitions of GIS used globally. [5]

- Q7)** a) Classify the Raster and Vector Data types. Explain with neat diagram. [6]
b) Differentiate various Attribute Data models and explain any one of them. [6]
c) Explain the application of GIS in Urban Planning. [6]

OR

- Q8)** a) Outline the Geo-referencing of GIS data? And explain its importance in the field of GIS? [6]
b) Enlist the applications of GIS in Civil Engineering & Explain any one in detail. [6]
c) Categorize the different Database? Name the types of database. [6]

